Venkata Lakshmi Parimala Pasupuleti

Senior Data Analyst · SQL · Python · Tableau · Power BI · ETL Pipelines · AWS · Generative AI

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McLean area, Virginia

PROFESSIONAL SUMMARY

Senior analyst with 8+ years building scalable self-service reporting solutions and data pipelines on high-stakes regulated datasets. At Entrata, built **property performance dashboards** tracking eligibility rates, occupancy, delinquency, and rent-vs-AMI trends across 100K+ unit portfolios; optimized **200+ SQL queries for 30% performance improvement** across 1M+ records; built ETL pipelines on **AWS (S3, RDS, Glue)** achieving 100% PHI compliance with zero incidents. At GMU, designed **self-service BI dashboards** (Tableau, ggplot2, Quarto) improving faculty stakeholder comprehension **40%** and built regression models achieving **$R^2 = 0.977$** across 190+ countries with automated statistical reporting. Championing GenAI adoption in analytics workflows: built **AgentProbe** (LLM-as-judge eval engine, multi-provider AI routing) and **Leasel** (agentic compliance pipeline, 93.6% precision). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — People Analytics · Statistical Modeling · Self-Service BI · Workday Reporting

- Built **self-service BI dashboards** (Tableau, ggplot2, Quarto) for faculty leadership translating complex model outputs into decision-ready visualizations — improved stakeholder comprehension **40%** across two cohorts; designed KPIs and performance metrics to evaluate research program effectiveness.
- Built end-to-end data modeling and ETL pipelines (Python, R) on multi-source datasets (WHO + UK Road Safety, 190+ countries) — regression models achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 pipelines surfacing statistical outliers manual review missed.
- Championed generative AI adoption in research analytics workflows — automated reporting pipelines (Python, Quarto) cutting manual analysis time; applied advanced statistical techniques including cross-validation, hypothesis testing, and PCA across multiple research projects.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Data Analytics · Scalable Reporting · AWS Data Infrastructure

- Built **property performance dashboards** across 100K+ unit portfolios — eligibility approval/denial rates, waitlist demand, occupancy and vacancy, rent burden vs. AMI, delinquency, and lease renewal trends; translated complex compliance data into actionable KPIs for stakeholders across operations and leadership teams.
- Migrated all client data to **AWS (S3, RDS, Glue)** with encrypted ETL pipelines — **100% PHI/HIPAA compliance**, zero security incidents; designed and shipped scalable data warehouse architecture supporting 1M+ property records at **99.5% uptime**.
- Optimized **200+ SQL queries for 30% performance improvement** across all client environments; built automated data quality and compliance validation pipelines eliminating **80% of manual review workload** with zero violations.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Compliance Analytics · Data Pipelines · Reporting

- Encoded manual income eligibility logic into automated analytics pipelines — AMI threshold calculations, recertification workflows, escalation routing across **millions of tenant certifications**; built audit-ready reports with full data lineage for compliance stakeholders.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack Development · SQL · Reporting Systems

- Built and maintained reporting features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; wrote and tuned queries across all client database environments supporting 100K+ unit portfolios.

PROJECTS

AgentProbe: GenAI Analytics Testing Toolkit

Python · LangGraph · FastAPI · ChromaDB · LLM-as-Judge · OpenAI · Anthropic · Groq · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a multi-agent AI platform with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) and multi-provider LLM routing — demonstrates practical generative AI integration into analytics and reporting workflows at production scale.

CrashClarity: Large-Scale ML Analytics (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble achieving **85%+ accuracy**; deployed interactive self-service dashboard on GitHub Pages for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ETL pipelines, data governance, ML analytics, data visualization, AI systems

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur
2013 – 2017

TECHNICAL SKILLS

Analytics & Modeling: Python, R, SQL, Statistical Modeling, Regression Analysis, Hypothesis Testing, A/B Testing, Isolation Forest, Random Forest, Logistic Regression, Feature Engineering, Cross-Validation, Scikit-learn, Pandas, NumPy

BI & Visualization: Tableau, Power BI, Workday Report Writer, ggplot2, Quarto, matplotlib, Jupyter, Self-Service Dashboards, KPI Design, Stakeholder Reporting, Data Storytelling

Data Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), ETL Pipelines, Data Pipelines, Data Modeling, Data Warehousing, PostgreSQL, MySQL, Oracle, SQLite, PHI/HIPAA Compliance, Audit Pipelines, Metadata & Lineage, Docker, CI/CD, Git, Linux

Generative AI & Automation: LangGraph, RAG Pipelines, LLM-as-Judge Evaluation, Multi-Provider LLM Routing, OpenAI API, Anthropic API, FastAPI, Automated Reporting Pipelines